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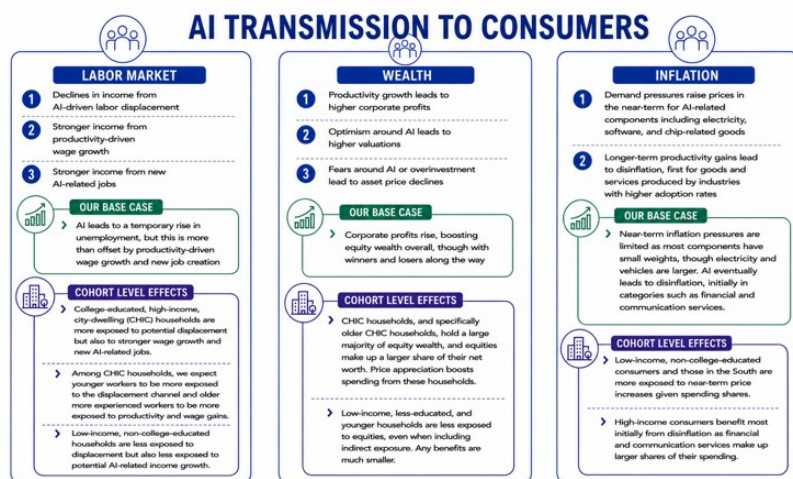
Charting AI's effects, from capex to consumption

The GDP contribution of AI infrastructure spend is already material, and in the coming years we expect a shift towards adoption spend. As adoption grows, effects across labor markets, asset markets, and prices will be key drivers of consumer spending and the outlooks for different consumer cohorts.

Key Takeaways

- We provide a more accurate treatment of AI investment's contribution to growth, including more investment categories and improving our treatment of imports.
- Broad AI-related spend contributed an avg of 0.6pp to real GDP growth since 2025. Isolating incremental "AI-only" spend points to an avg contribution of 0.4pp.
- Of the roughly 0.5pp contribution from AI-only investment to growth in 1H26, about 0.4pp came from infrastructure, with the remainder from adoption.
- High-income, college-educated households are most exposed to displacement but also to upside from wage growth, job creation, wealth effects, and disinflation.
- In our base case, these affluent households reap the largest gains as upside channels outweigh downside risks, but diffusion speed will be an important factor.

Exhibit 1: Overview of AI transmission to consumers



Source: Morgan Stanley Research

MORGAN STANLEY & CO. LLC

Michael T Gapen

Chief US Economist

Michael.Gapen@morganstanley.com

+1 212 761-0571

Sam D Coffin

Economist

Sam.Coffin@morganstanley.com

+1 212 761-4630

Diego Anzoategui

Economist

Diego.Anzoategui@morganstanley.com

+1 212 761-8573

Arunima Sinha

Global Economist

Arunima.Sinha@morganstanley.com

+1 212 761-4125

Heather Berger

Economist

Heather.Berger@morganstanley.com

+1 212 761-2296

Lingdi Xu

Economist

Lingdi.Xu@morganstanley.com

+1 212 761-2957

Charting AI's effects, from capex to consumption

Refining our estimate of AI's contribution to GDP

Understanding the US outlook increasingly requires understanding where AI is showing up in the economy. On the investment side, our updated framework suggests the contribution is already material. In our recent note, [Building a better mousetrap: AI spending and its contribution to US GDP growth](#), we refined our estimate of AI investment's contribution to GDP. In our new estimates, we broadened the set of investment categories included, improved the treatment of imports, and distinguished between AI-related spending and spending that is incremental to the AI boom.

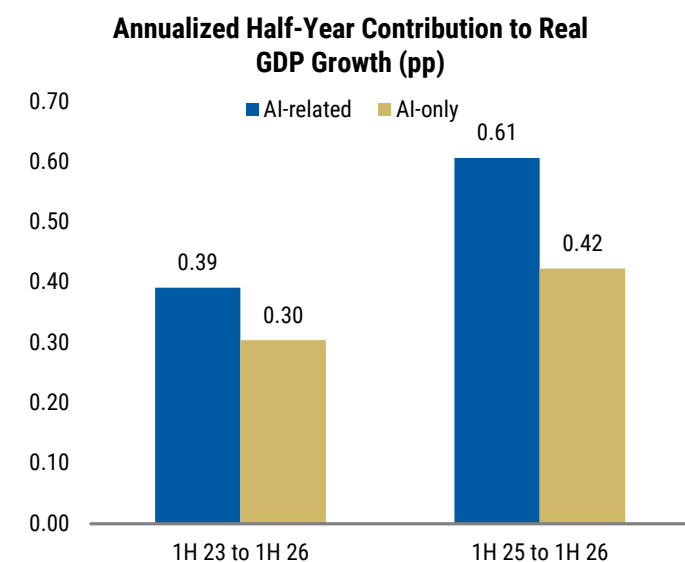
Translating AI spending into growth is difficult since capex does not translate one-for-one into US GDP. GDP captures domestically produced final output, whereas corporate capex can include imported and intermediate goods. In the case of AI investment, spending on imported equipment used in data center construction is included in corporate capex plans but does not contribute to US GDP.

Using our broad measure, AI-related investment has contributed about 0.6pp on average to annualized real GDP growth since 2025; the narrower incremental measure, which we call "AI-only" spending, puts the contribution closer to 0.4pp. Our new treatment of imports raises our estimate of the growth impulse versus our prior calculations, as we adjust for re-exports and also estimate the share of computer imports being used for personal consumption instead of business investment.

In addition to the total growth effects, we also separate out AI infrastructure spend—data centers, computing equipment, networking hardware, power infrastructure and related machinery—from AI adoption, for which software investment is our best economy-wide proxy. So far, infrastructure still dominates. Of the roughly 0.5pp contribution from AI-only investment to growth in 1H26, about 0.4pp came from infrastructure, with the remainder from adoption. The infrastructure contribution has accelerated from about 0.15pp in 2023 and -0.05pp in 2024 to 0.21pp in 2025 and 0.36pp in 1H26.

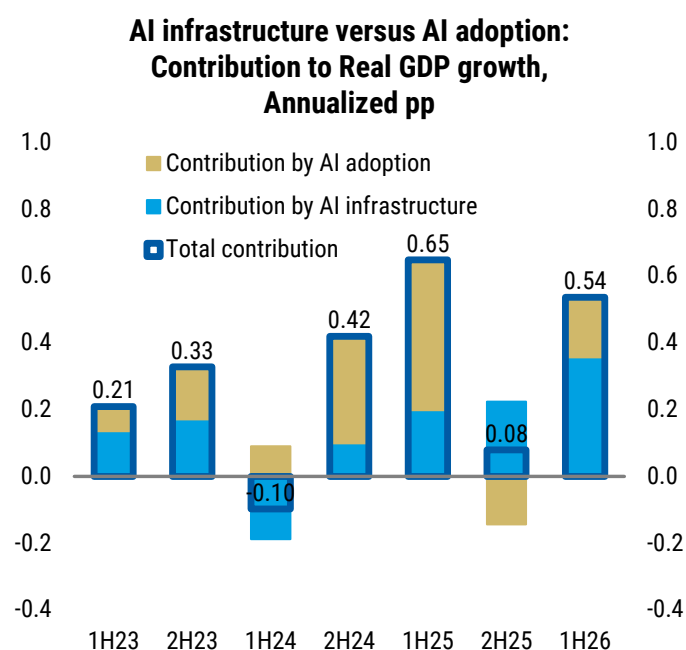
We expect the mix to evolve as the cycle matures. Infrastructure spending is creating the supply-side capacity needed to develop and run AI applications; software spending should indicate whether AI is diffusing across the broader business sector. Adoption has picked up, but not decisively: software contributed about 0.18pp to growth in 2026, after 0.15pp in 2025 and 0.21pp in 2024. Our expectation is that infrastructure growth slows in 2027-28 as the build-out matures, while adoption spending proves more persistent and potentially accelerates.

Exhibit 2: The contribution of AI investment to growth in real US GDP accelerated beginning in 2025



Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 3: Decomposing the contribution of AI-only spending to growth into its infrastructure and adoption subcomponents



Source: BEA, Haver Analytics, Morgan Stanley Research

AI's effects across consumer cohorts

The shift from infrastructure spend to adoption will affect not only business investment, but also personal consumption. So far, most discussions of AI's distributional effects have focused on the labor disruption channel. While occupational exposure is important, it gives an incomplete answer. In our recent note, [Dispelling an AI Myth: Why Affluent Consumers May Be Safer Than Advertised](#), we assessed AI exposure across consumer cohorts through income, wealth, and prices, and discussed how these channels net out in different scenarios.

Starting with the labor channel, high-income, college-educated, city-dwelling—"CHIC"—households are disproportionately represented in highly AI-exposed occupations, but exposure can mean either automation or augmentation. While [we have seen](#) larger increases in unemployment in high AI exposure occupations, displacement has so far been more extreme for young workers. For experienced workers, AI could augment tasks, boosting productivity and wages. Meanwhile, new AI-related jobs have so far been aimed at these same high-income, highly educated consumers, providing an important offset.

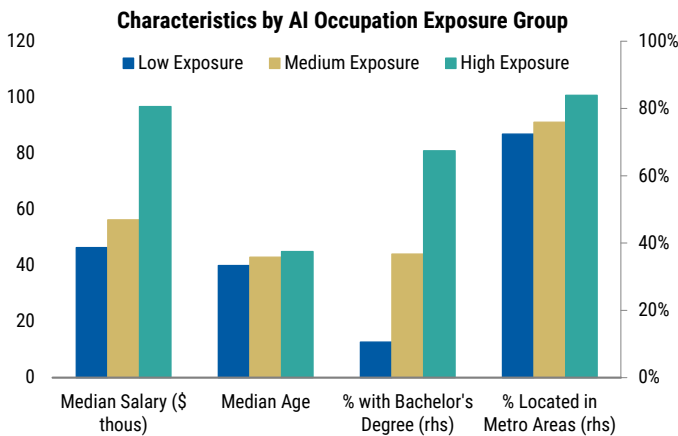
Wealth is the second channel and is especially important for these same CHIC consumers. The top 20% income cohort, college-educated group, and 55+ cohort each account for around 80% of household equity wealth. For the top income cohort, equity wealth is roughly six times annual labor income, so relatively modest asset-price gains can offset the spending effects of a temporary decline in labor income. For lower-income consumers, equity wealth is lower both in dollars and as a share of their own wealth, so equity moves have less of an impact on spending.

Finally, changes in income and wealth must be considered alongside changes in prices. In

the near term, strong demand associated with the AI build-out can put upward pressure on electricity, software and chip-intensive goods. Many of these components are relatively small and do not derail our near-term expectations for disinflation, but this AI-exposed basket makes up a larger share of spending for low-income, less-educated, and younger households. Over the longer run, productivity gains should be disinflationary, particularly where adoption is faster. Higher-income households spend a larger share on financial, professional and communication services, so they may experience AI-related services disinflation earlier.

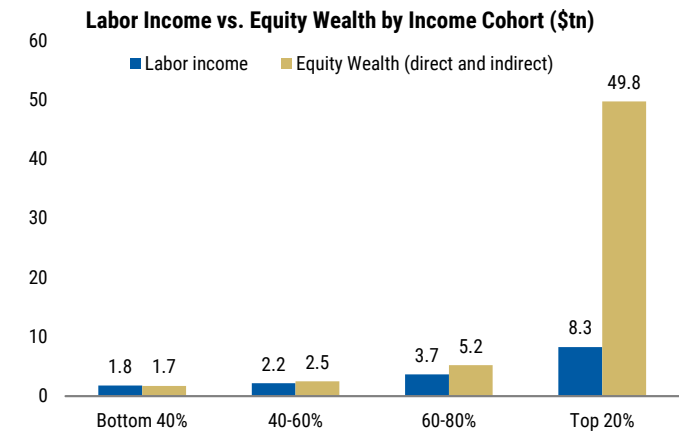
Putting the channels together, [our base case remains](#) that AI will lead to a modest, temporary increase in unemployment (around 0.2-0.5pp at the peak), but that task creation, wage gains, and wealth effects will more than offset this displacement. In this scenario, CHIC households benefit the most on net, as upside channels outweigh downside risks. Still, outcomes likely vary by career stage. Established, older professionals may be well positioned to capture AI-related productivity and asset-price gains while highly educated younger professionals may be more vulnerable. Weakness among younger entrants would likely have a smaller effect on aggregate consumption than a comparable disruption among established households that account for a large share of spending. The key risk is a diffusion cycle that moves materially faster than workers and firms can adjust. In that bear case, displacement first hits highly exposed white-collar households, but the demand shock eventually broadens the weakness to other cohorts, and monetary and fiscal responses will be key.

Exhibit 4: High-exposure AI occupations tend to be higher-income, more educated and concentrated in cities



Source: IPUMS, Morgan Stanley Research

Exhibit 5: For high-income consumers, equity wealth is very elevated relative to labor income



Source: BLS, Federal Reserve Board, Haver Analytics, Morgan Stanley Research

Oil tracker: Elevated prices, falling inventories

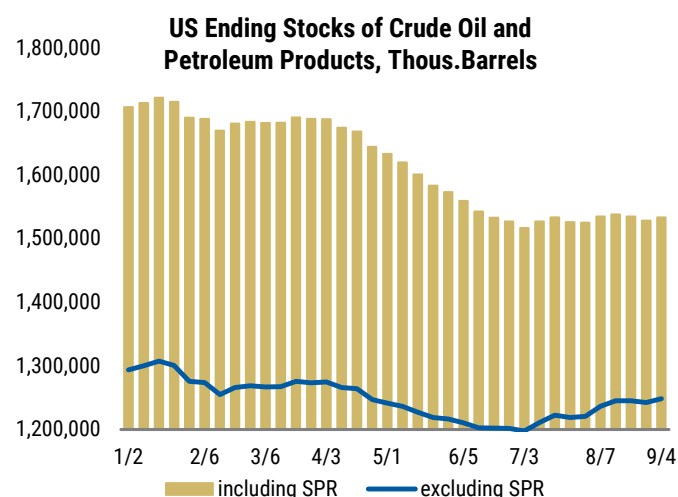
We continue to track the evolution of petroleum-product inventories, production, trade, and prices. The latest data suggest that crude oil inventories in the Strategic Petroleum Reserve (SPR) fell further, and the total US crude oil and petroleum-product inventories were relatively flat. US domestic crude oil production rose, while net imports of crude oil and petroleum products have moved sideways recently. Meanwhile, oil prices moved higher.

As of September 4, US stocks of crude oil and petroleum products, which represent the EIA's estimate of how many barrels of crude oil and petroleum products are physically sitting in US storage at period-end, remained broadly unchanged from a week earlier. The stocks of crude oil in the Strategic Petroleum Reserve (SPR) continued to fall to around 285 million barrels, the lowest level since April 1984. The Energy Department said the SPR's minimum inventory is determined by "cavern mechanics," which, when applied across the full system, translates to a conservative operational minimum of about 70 million barrels. If the SPR continues to decline at roughly its recent pace of 30 million barrels every four weeks, inventories would approach the operational minimum by the end of February 2027.

Following the recent re-escalation between the US and Iran, both spot and futures prices have risen, renewing concerns about inflation. The latest spot prices, as of September 8, were \$94.21 per barrel for WTI in Cushing, Oklahoma, and \$106.12 per barrel for Brent in Europe.

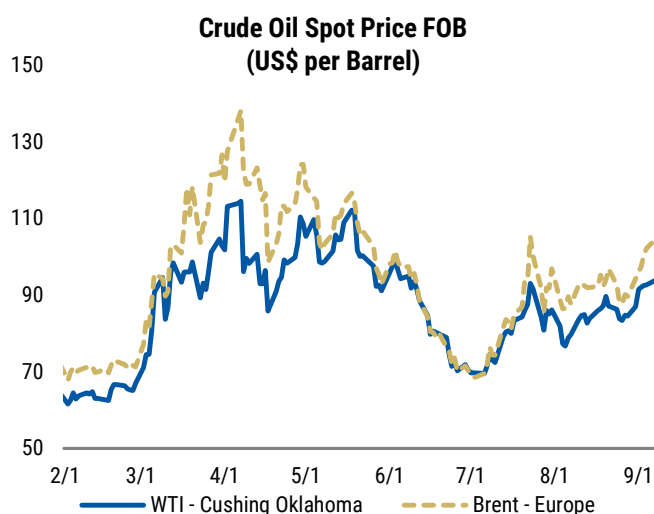
US domestic crude production has been trending up since early May. Net imports of oil and petroleum products (imports minus exports) have been trending up from their low in April as crude oil exports have gradually trended down since then.

Exhibit 6: US ending stocks of crude oil and petroleum products edged up recently



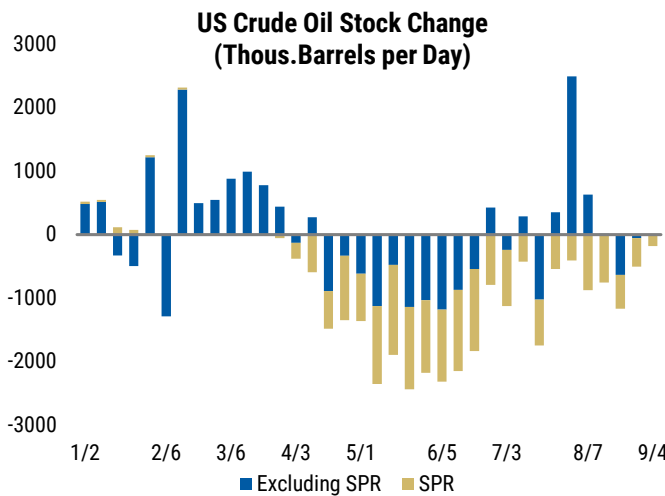
Source: Energy Information Administration, Haver Analytics, Morgan Stanley Research

Exhibit 7: Following the recent re-escalation, the spot price of oil for immediate physical delivery has risen meaningfully



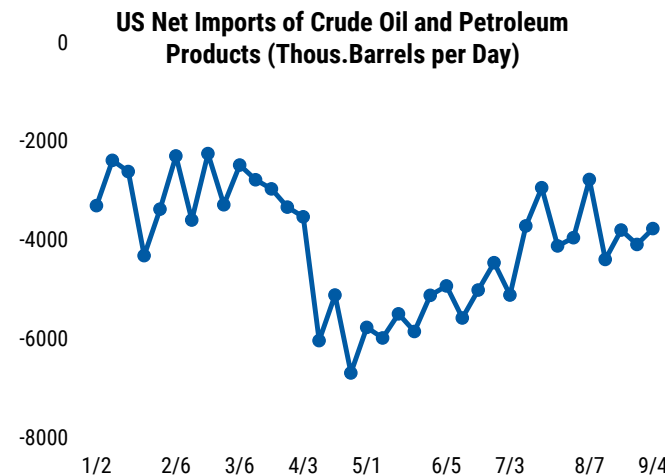
Source: Energy Information Administration, Haver Analytics, Morgan Stanley Research

Exhibit 8: Including the SPR, US oil stocks are now falling less than 1mn barrels per day



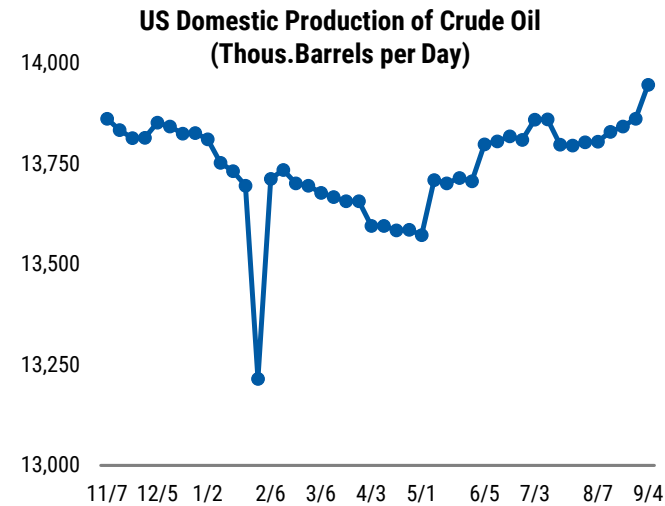
Source: Energy Information Administration, Haver Analytics, Morgan Stanley Research

Exhibit 10: US exports of crude oil trended down over the past two months, pushing up net imports of crude oil and petroleum products



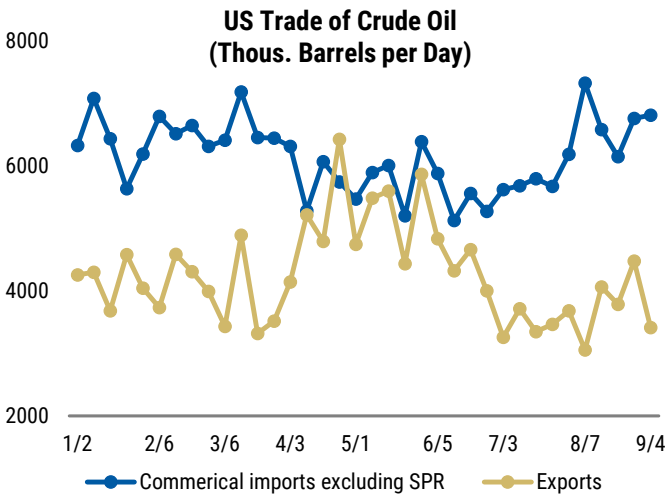
Source: Energy Information Administration, Haver Analytics, Morgan Stanley Research

Exhibit 9: US domestic oil production has been rising



Source: Energy Information Administration, Haver Analytics, Morgan Stanley Research

Exhibit 11: US exports of oil decelerated following the signing of the MOU



Source: Energy Information Administration, Haver Analytics, Morgan Stanley Research

Financial conditions: No longer easing

Following the conflict in the Middle East and the uncertainty it brings to the next steps for monetary policy, we continue to track our FRB/US-based measure of financial conditions.

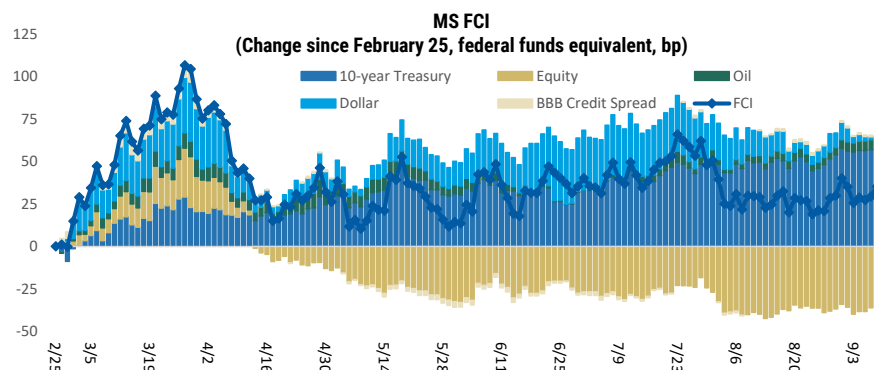
As of the September 10 market close, financial conditions remained tighter than before the recent escalation in the Middle East, and no longer easier than at the time before the July FOMC meeting. Since the July FOMC meeting, financial conditions eased and reversed, now broadly unchanged.

The index aims to capture how changes in asset prices weigh on future economic activity. It includes five daily variables: the 10-year Treasury yield, S&P 500 returns, the corporate BBB credit spread, the valuation of the U.S. dollar, and the price of oil. These are aggregated based on their estimated growth elasticities relative to the federal funds rate using the Federal Reserve's FRB/US model. As a result, the index can be interpreted as the equivalent change in the federal funds rate, expressed in basis points, required to generate a similar effect on economic activity.

Since hostilities in the Middle East began on February 28, the tightening in financial conditions is equivalent to about a 53bp rise in the federal funds rate. This has reversed the easing seen earlier in the year, most of which was driven by a weaker U.S. dollar. The net tightening since February 28 has been driven mainly by higher 10-year U.S. Treasury yields and elevated oil prices.

The contribution from oil prices disappeared for a couple of weeks following the MOU between Iran and the U.S. but has now reappeared following the recent re-escalation. Meanwhile, 10-year Treasury yields have continued to move higher, contributing to tighter financial conditions, while buoyant equity markets have offset some of the overall tightening.

Exhibit 12: Financial conditions have tightened following the conflict in the Middle East



Source: Bloomberg, Morgan Stanley Research

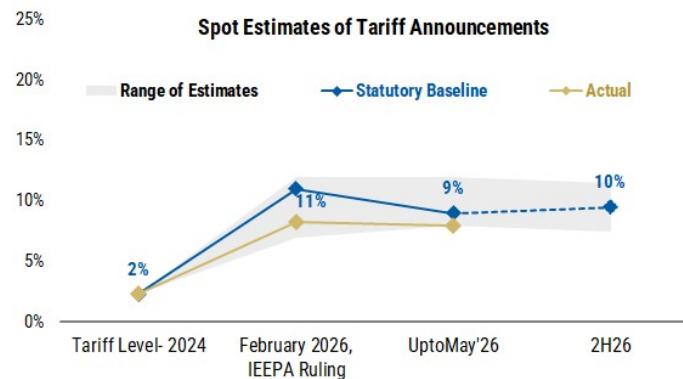
The effective tariff rate, tariff receipts, and refunds

Tariffs averaged 6.7% over May-July, and we expect that it will rise to near 10%

USTR's final forced-labor Section 301 action was broadly in line with our expectations.

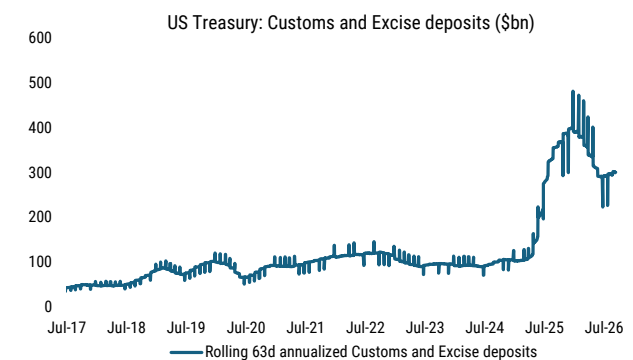
Effective July 24, the agency imposed additional tariffs on imports from 60 economies, setting a 10% duty on economies that have adopted, partially adopted, or committed to adopt a forced-labor import prohibition and a 12.5% duty on most others. The final action also preserves several key carve-outs, including goods already subject to Section 232 tariffs, USMCA-compliant imports, and a range of general and country-specific product exemptions. Overall, both the tariff structure and the principal exemptions closely resemble the proposal, reinforcing our expectation that Section 301 would serve as the broad country-level tariff floor. We continue to expect that the statutory effective tariff rate will converge toward ~10% by year-end. In the July 2026 data, the effective tariff rate is estimated at ~6.5%. Over the past three months of detailed trade data (May, June and July), the tariff rate has averaged 6.7%. There has been a recent escalation of trade disputes with Canada, and [section 338 tariffs have been applied on approximately \\$20bn of Canadian imports](#). However, for now, we do not expect this to meaningfully affect our year-end tariff estimates.

Exhibit 13: US effective tariff rate



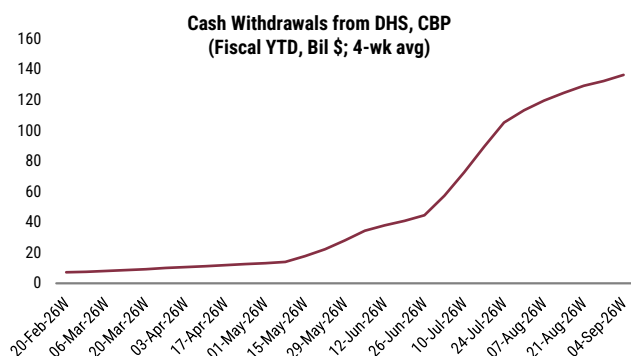
Source: US Census, US HTS, USITC, Morgan Stanley Research. The "Up to May '26" estimate is the average tariff rate on all imports over Jan - May 2026.

Exhibit 14: US Treasury: Customs and excise deposits from tariffs



Source: US Treasury, Bloomberg, Morgan Stanley Research

Exhibit 15: Cash withdrawals from the DHS - CBP, as a proxy for tariff refunds



Source: US Treasury, Haver Analytics, Morgan Stanley Research

Tariff refunds represent cash returned to importers when customs duties were previously overpaid or later determined to be refundable, for different reasons, including a tariff exclusion or court-ordered refund or a drawback claim tied to goods that are exported or destroyed. The process generally runs through CBP, where importers or brokers submit the relevant claim and supporting documentation; CBP then reviews eligibility, confirms the entry and status, and certifies approved refunds for payment. In Treasury cash-flow data, we monitor these payments at high frequency through CBP-related withdrawals in the Daily Treasury Statement. We interpret this as a tariff-refund proxy.

3Q GDP tracking at 2.0%

Our 3Q GDP tracking fell a tenth to 2.0% in the latest week after existing home sales lowered our residential investment estimate. The Atlanta Fed's estimate, much higher than ours, was unchanged at 4.7%. Its goods consumption estimate, at 2.3%, is 2½ points above ours; its services estimate, at 4.5%, is 1½ point above ours. It also allows for much faster inventory investment than we do. The New York Fed nowcast moved up a tenth to 2.3%, mostly reflecting strong August payrolls.

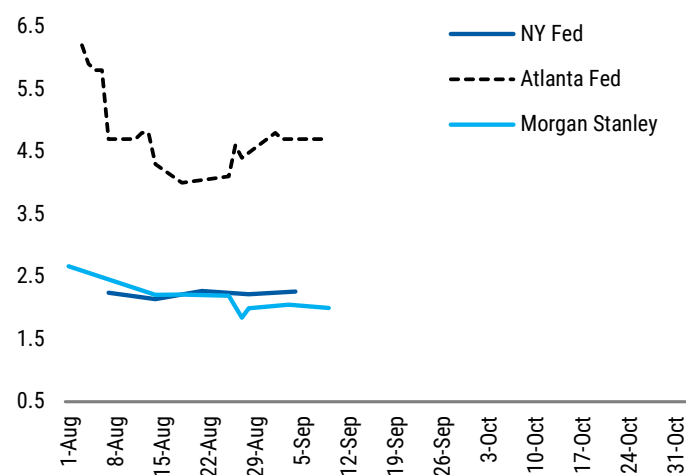
Exhibit 16: The effect of incoming data on our US GDP tracking estimate

Details of Q3 2026 US GDP tracking (% q/q saar unless indicated)												
Date	Data release	GDP	Final sales	PCE	Res	Equip	Struct	IPP	Gov	X	M	NI (level)
Aug 7	Baseline	2.7	2.6	2.4	1.0	8.0	-3.0	8.0	1.4	1.0	2.0	-1086
Aug 14	Retail sales and CPI	2.2	2.1	1.8	1.0	8.0	-3.0	8.0	1.4	1.0	2.0	-1086
Aug 18	Housing starts and permits	2.2	2.1	1.9	-1.5	8.0	-3.0	8.0	1.4	1.0	2.0	-1086
Aug 25	New home sales	2.2	2.1	1.9	-2.0	8.0	-3.0	8.0	1.4	1.0	2.0	-1086
Aug 26-27	Trade and durables	1.8	1.6	1.9	-2.0	8.0	-3.0	8.0	1.5	-3.3	2.4	-1130
Aug 28	2Q consumption	2.0	1.7	2.1	-2.0	8.0	-3.0	8.0	1.5	-3.3	2.4	-1130
Sep 3	Trade detail	2.1	1.6	2.1	-2.0	15.0	-3.0	8.0	1.5	-6.0	4.0	-1165
Sep 10	Existing home sales	2.0	1.6	2.1	-2.3	15.0	-3.0	8.0	1.5	-6.0	4.0	-1165
GDP tracking		2.0	1.6	2.1	-2.3	15.0	-3.0	8.0	1.5	-6.0	4.0	-1165
Contribution to GDP growth (pp)		1.6	1.5	-0.1	0.9	-0.1	0.5	0.2	-0.7	-0.6	-1.3	0.4
Morgan Stanley official forecast		2.7	2.6	2.4	1.0	8.0	-3.0	8.0	1.4	1.0	2.0	-1086

Note: Our GDP tracking estimate is distinct from our official published GDP forecast. It reflects the mechanical aggregation of monthly activity data that feed directly into the BEA's calculation. Where data releases have implications for the tracking components, cells have been bolded. Final sales is GDP less inventories. Source: Morgan Stanley Research

Exhibit 17: GDP nowcasts

Real GDP tracking, Q/Q % change, a.r.



Source: Federal Reserve Banks of New York and Atlanta, Morgan Stanley Research

Exhibit 18: Our tracking vs. the Atlanta Fed tracking

	Morgan Stanley		Atlanta Fed		Difference
	q/q%, ar	contrib	q/q%, ar	contrib	
GDP	2.0		4.7		-2.7
Final sales	1.6	1.6	2.7	2.7	-1.1
Final sales to dom. purchasers	2.8	2.9	4.0	4.1	-1.3
Private	3.0	2.6	4.6	4.0	-1.4
Consumption	2.1	1.5	3.8	2.6	-1.1
Goods	0.0	0.0	2.3	0.5	-0.5
Services	3.1	1.5	4.5	2.1	-0.6
Residential investment	-2.3	-0.1	-2.3	-0.1	0.0
Equipment investment	15.0	0.9	18.3	1.0	-0.1
Nonres structures	-3.0	-0.1	2.7	0.1	-0.2
Intel. Property	8.0	0.5	6.8	0.4	0.1
Government	1.5	0.2	0.9	0.2	0.1
Exports	-6.0	-0.7	-2.1	-0.3	-0.4
Imports	4.0	-0.6	8.6	-1.2	0.6
Net exports		-1.3		-1.5	0.2
Inventories		0.4		2.1	-1.6

Source: Federal Reserve Bank of New York, Morgan Stanley Research

Data review & preview

Review of past week's data

Exhibit 19:

Monday 07 Sept		Data	Period	Forecast	Consensus	Prev	Actual
		Labor Day: No scheduled data or Fedspeak					
Tuesday 08 Sept		Data	Period	Forecast	Consensus	Prev	Actual
6:00 AM		NFIB Small Business Optimism	Aug		99.2	99.8	98.7
11:00 AM		NY Fed 1-Yr Inflation Expectations	Aug			3.6	3.6
3:00 PM		Consumer Credit	Jul		11.3	14.2	18.1
Wednesday 09 Sept		Data	Period	Forecast	Consensus	Prev	Actual
7:00 AM		MBA Mortgage Applications	4-Sep				-2.7
8:15 AM		ADP Weekly Employment Change	22-Aug			11.8	10.0
10:00 AM		Quarter Services Survey	2Q F				
Thursday 10 Sept		Data	Period	Forecast	Consensus	Prev	Actual
8:30 AM		Initial Jobless Claims	5-Sep	200	205.0	206	206
8:30 AM		Continuing Claims	29-Aug	1780		1779	1774
8:30 AM		PPI Final Demand m/m	Aug		0.4	0.0	0.4
8:30 AM		PPI Ex Food, Energy, Trade m/m	Aug		0.3	0.4	0.3
8:30 AM		PPI Final Demand y/y	Aug		5.2	4.7	5.4
8:30 AM		PPI Ex Food, Energy, Trade y/y	Aug			4.7	4.7
10:00 AM		Existing Home Sales	Aug	3.97	3.99	4.06	4.0
10:00 AM		Existing Home Sales m/m	Aug	-2.2	-1.6	-1.7	-2.0
10:00 AM		Wholesale Inventories m/m	Jul F			1.3	1.3
4:30 PM		Fed balance sheet (H.4.1)					
Friday 11 Sept		Data	Period	Forecast	Consensus	Prev	Actual
8:30 AM		CPI m/m	Aug	0.41	0.4	0.1	
8:30 AM		Core CPI m/m	Aug	0.23	0.2	0.2	
8:30 AM		CPI y/y	Aug	3.37	3.4	3.4	
8:30 AM		Core CPI y/y	Aug	2.38	2.4	2.5	
8:30 AM		CPI Index NSA	Aug	335.077	334.957	333.918	
10:00 AM		U. of Mich. Sentiment	Sep P		51.0	51.7	
10:00 AM		U. of Mich. 5-10 Yr Inflation	Sep P			3.3	
12:00 PM		Household Change in Net Worth	2Q			113.1	
2:00 PM		Federal Budget Balance	Aug			-344.8	
4:15 PM		Commercial bank balance sheets (H.8)					

Source: Bloomberg, Morgan Stanley Research forecasts

Preview of upcoming data

Exhibit 20: Monday

Monday 14 Sept		Data	Period	Forecast	Consensus	Prev
		No scheduled data or Fedspeak				

Source: Bloomberg, Morgan Stanley Research forecasts

Exhibit 21: Tuesday

Tuesday 15 Sept		Data	Period	Forecast	Consensus	Prev
8:15 AM		ADP Weekly Employment Change	29-Aug			12.0
8:30 AM		Empire Manufacturing	Sep		12.1	20.6

Source: Bloomberg, Morgan Stanley Research forecasts

Exhibit 22: Wednesday

Wednesday 16 Sept	Data	Period	Forecast	Consensus	Prev
7:00 AM	MBA Mortgage Applications	11-Sep			-2.7
8:30 AM	New York Fed Services Business Activity	Sep			0.5
8:30 AM	Import Price Index m/m	Aug			-0.4
8:30 AM	Export Price Index m/m	Aug			-1.3
8:30 AM	Retail Sales Advance m/m	Aug	0.7	0.9	-0.6
8:30 AM	Retail Sales Ex Auto m/m	Aug	0.5	0.5	-0.3
8:30 AM	Retail Sales Control Group	Aug	0.6	0.4	-0.4
10:00 AM	NAHB Housing Market Index	Sep		34.0	35.0
10:00 AM	Business Inventories	Jul		0.2	0.0
2:00 PM	FOMC Rate Decision (Upper Bound)	16-Sep		3.8	3.8
4:00 PM	Total Net TIC Flows	Jul			133.5

Source: Bloomberg, Morgan Stanley Research forecasts

FOMC decision. Fed Chairman Warsh delivered a hawkish Jackson Hole speech. However, we think inflation data will determine the Fed's next move, and that evidence of further disinflation in Friday's CPI for August will support a hold. Expected downward revisions to PCE inflation could provide an additional argument for patience. Since Jackson Hole, a few FOMC officials—Barr, Waller, Williams—have suggested that recent disinflation, if it extends, could be consistent with a hold rather than a hike.

Retail sales. We expect total retail sales rose 0.7%*m/m* in August, with retail control group sales rising 0.6%*m/m*. Within total retail sales, we expect auto sales were strong (up 1.6%*m/m* in August), and sales at gasoline stations picked up again (up 0.2%*m/m*) as gas prices rose. Within retail control, several factors boost our forecast: monthly gains in nominal labor market income, a recovery in the weak online sales, and continued support from back-to-school activity. Although Labor Day fell well into September this year, there was early promotional activity, which likely contributed to goods sales. The seasonal factor and control group prices are not significant drivers in our forecast for the retail control group. In other components, we expect that sales at restaurants were up in August (0.2%*m/m*) and sales of building materials showed a dip (down 0.2% *m/m*).

Import prices. We will finalize our core PCE tracking (currently at 0.24%*m/m* before the CPI release) incorporating international airfares prices from the import prices release. Import prices *y/y* inflation rose steadily until May, and it is still at elevated levels, reflecting pressures from the conflict (through oil and materials) and the AI demand push (through computers, peripherals, and semiconductors).

Exhibit 23: Thursday

Thursday 17 Sept	Data	Period	Forecast	Consensus	Prev
8:30 AM	Philadelphia Fed Business Outlook	Sep		28.6	47
8:30 AM	Initial Jobless Claims	12-Sep	200		206
8:30 AM	Continuing Claims	5-Sep	1780		1774
8:30 AM	Housing Starts	Aug	1365	1318	1239
8:30 AM	Housing Starts <i>m/m</i>	Aug	10.2	6.9	-12.4
8:30 AM	Building Permits	Aug P	1425	1413	1433
8:30 AM	Building Permits <i>m/m</i>	Aug P	-1.2		4.3
10:00 AM	Pending Home Sales <i>m/m</i>	Aug			-2.3

Source: Bloomberg, Morgan Stanley Research

Jobless claims. We expect new jobless claims hover around 200k in the weeks

surrounding Labor Day. Both new and continuing claims are below their levels of a year ago. Layoffs have slowed further, but hiring too remains very slow.

Housing starts and building permits. We expect single-family starts rebound from last month's soft print, +13% m/m to 915k, as permits and the Housing Market Index did not show the same weakness. This estimate assumes there is also some upward revision to July's previously soft print. If not, the rebound could likely be larger. We expect single-family permits unchanged at 895k. On the multifamily side, we expect starts rise 4.4% m/m to 450k while permits decline 3.5% to 530k.

Exhibit 24: Friday

Friday 18 Sept	Data	Period	Forecast	Consensus	Prev
9:15 AM	Industrial Production m/m	Aug	0.3	0.3	0.2
9:15 AM	Manufacturing Production m/m	Aug	0.3		0.2
9:15 AM	Capacity Utilization	Aug		76.5	76.3
10:00 AM	Leading Index	Aug		0.2	0.2

Source: Bloomberg, Morgan Stanley Research

Industrial production probably rose for the seventh month in the last nine, with broad gains in nonauto manufacturing and further increase in utilities output. Nonauto manufacturing output has increased by 1.1% over the past year, with broad gains across durable goods industries. Within durables (+3.9% y/y), there has been particular strength in computers (+9.9%), electrical equipment (+6.2%), and aerospace (4.5%). Nondurables output has fallen 1.6% y/y. Utilities output has increased by 0.7% y/y, as AI-related demand appears to have driven commercial usage to 4.5% growth.

By industry, it's Capex over Consumption, as consumer goods output has declined 1.8% y/y while business equipment output has risen 6.6% and defense equipment has risen 6.8%. None of these patterns seem likely to change, with AI continuing to drive demand for capital goods and electricity, and with strength in demand for defense goods. And with strength in capital goods demand, there is also accelerating output of nonindustrial supplies (construction supplies, general business supplies, energy) and in durable goods materials (computer parts, semiconductors, etc).

Data calendar

Exhibit 25: September 14 - September 25

<i>Monday</i>	<i>TUESDAY</i>	<i>WEDNESDAY</i>	<i>THURSDAY</i>	<i>FRIDAY</i>
14	15	16	17	18
	ADP Weekly Employment Change (Wk of 8/29, 8:15 AM) Empire Manufacturing (Sep, 8:30 AM) 20-Year Bond Auction (1 PM) Treasury Buyback (TIPS 10Y-30Y, max \$500 mil, 1:40 PM)	MBA Mortgage Applications (Wk of 9/11, 7 AM) NY Fed Services Business Activity (Sep, 8:30 AM) Import & Export Prices (Aug, 8:30 AM) Retail Sales (Aug, 8:30 AM) NAHB Housing Market Index (Sep, 10 AM) Business Inventories (Jul, 10 AM) FOMC Rate Decision (2 PM) TIC data (Jul, 4 PM)	Philadelphia Fed Manufacturing (Sep, 8:30 AM) Jobless Claims (Wk of 9/12, 8:30 AM) Housing Starts & Building Permits (Aug, 8:30 AM) Pending Home Sales (Aug, 10 AM) 10-Year TIPS Auction (1 PM) Treasury Buyback (Nom Cpn 7Y-10Y, max \$4 bil, 1:40 PM)	Industrial Production (Aug, 9:15 AM) Leading Index (Aug, 10 AM)
21	22	23	24	25
Fed's Goolsbee Speaks (6:30 AM) Chicago Fed Nat Activity Index (Aug, 8:30 AM)	ADP Weekly Employment Change (Wk of 9/5, 8:15 AM) Philadelphia Fed Non-Manufacturing Activity (Sep, 8:30 AM) Richmond Fed Manuf. Index (Sep, 10 AM) 2-Year Notes Auction (1 PM)	MBA Mortgage Applications (Wk of 9/18, 7 AM) S&P Flash Manuf. (Sep P, 9:45 AM) S&P Services PMI (Sep P, 9:45 AM) 2-Year FRN Auction (11:30 AM) 5-Year Notes Auction (1 PM)	Current Account Balance (2Q, 8:30 AM) Jobless Claims (Wk of 9/19, 8:30 AM) New Home Sales (Aug, 10 AM) Fed's Paulson speaks (10:10 AM) Kansas City Fed Manf. Activity (Sep, 11 AM) 7-Year Notes Auction (1 PM) Treasury Buyback (Nom Cpn 20Y-30Y, max >=\$4 bil, 1:40 PM)	Bloomberg Sept. United States Economic Survey (6 AM) Durable Orders (Aug P, 8:30 AM) U. of Mich. Consumer Sentiment (Sep F, 10 AM) Kansas City Fed Services Activity (Sep, 11 AM)

Note: For updates post-publication of the Treasury Auction Schedule, see [here](#). Source: Bloomberg, Morgan Stanley Research

US economic outlook

Exhibit 26: Morgan Stanley forecasts

	4Q/4Q % change			Quarterly % change, annual rate (unless otherwise noted)									
	2025A	2026E	2027E	3Q25A	4Q25A	1Q26A	2Q26A	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E
Real GDP	2.0	2.2	2.6	4.4	0.5	2.1	1.5	2.7	2.5	2.4	2.6	2.6	2.6
Final Sales ¹	2.2	2.3	2.3	4.5	0.3	1.9	2.2	2.6	2.4	2.1	2.3	2.3	2.3
Final Domestic Demand²	1.8	2.6	2.8	2.8	0.6	2.2	3.3	2.7	2.3	2.6	2.8	2.8	2.8
Final Private Domestic Demand ³	2.4	2.8	3.1	2.9	1.8	1.7	4.2	2.9	2.5	2.8	3.1	3.2	3.2
Personal Consumption Expenditures	2.1	2.1	2.1	3.5	1.9	0.5	3.4	2.4	1.9	1.8	2.2	2.2	2.2
– Goods	1.4	2.2	1.6	3.0	0.3	0.5	4.3	3.2	1.0	1.2	1.7	1.7	1.7
– Services	2.4	2.0	2.3	3.6	2.7	0.5	3.1	2.1	2.3	2.1	2.4	2.4	2.4
Nonresidential Fixed Investment	5.6	7.7	8.0	3.2	2.4	10.6	8.5	5.9	5.9	7.9	7.9	8.0	8.0
– Structures	-5.5	-3.1	2.0	-5.0	-6.5	-4.7	-1.8	-3.0	-3.0	2.0	2.0	2.0	2.0
– Equipment	9.6	11.3	10.5	5.2	4.3	15.8	13.6	8.0	8.0	10.5	10.5	10.5	10.5
– IPP	8.0	9.6	8.0	5.6	5.4	13.8	8.8	8.0	8.0	8.0	8.0	8.0	8.0
Residential Investment	-3.8	-1.2	1.5	-7.1	-1.7	-7.8	1.3	1.0	1.0	1.5	1.5	1.5	1.5
Exports	1.1	4.7	1.8	9.6	-3.2	10.9	4.5	1.0	2.5	1.8	1.8	1.8	1.8
Imports	-1.9	7.1	5.0	-4.4	-1.0	11.8	12.5	2.0	2.0	5.0	5.0	5.0	5.0
Government	-1.2	1.6	1.3	2.2	-5.6	4.4	-1.0	1.4	1.4	1.3	1.3	1.3	1.3
Inventory contribution (pct pts, a.r.)	-0.2	-0.1	0.3	-0.1	0.1	0.2	-0.7	0.0	0.0	0.2	0.2	0.2	0.2
Trade contribution (pct pts, a.r.)	0.4	-0.4	-0.4	1.6	-0.2	-0.4	-1.1	-0.2	0.0	-0.5	-0.5	-0.6	0.0
Nominal GDP (Current \$)	5.4	5.8	4.8	8.3	4.2	5.8	8.0	4.9	4.7	4.9	4.5	4.6	4.9
Nominal Consumption	5.0	5.7	4.2	6.3	4.9	5.2	8.9	4.7	4.0	4.5	3.9	4.0	4.5
Employment & Personal Income													
Civilian Unemployment Rate (%)	4.5	4.3	4.1	4.3	4.5	4.3	4.3	4.1	4.3	4.3	4.2	4.2	4.1
Civilian Labor Force Participation Rate (%)	62.5	61.4	61.2	62.3	62.5	62.0	61.7	61.4	61.4	61.3	61.3	61.3	61.2
Employment to Population Ratio (%)	59.7	58.8	58.7	59.6	59.7	59.3	59.1	58.9	58.8	58.7	58.7	58.7	58.7
Average Monthly Change in Nonfarm Payrolls (Thous.)	10	57	59	23	-39	73	77	32	45	57	60	60	60
Real DPI	1.0	1.0	1.6	1.0	-0.9	0.9	-1.5	2.0	2.8	1.1	1.0	1.8	2.4
Saving rate (%)	3.8	2.9	2.5	4.4	3.8	3.9	2.8	2.7	2.9	2.8	2.5	2.4	2.5
Business Indicators													
Industrial Production	1.5	2.0	2.5	2.1	-2.0	1.3	4.1	1.3	1.1	2.4	2.5	2.5	2.5
Productivity	2.5	1.4	2.1	5.2	1.6	0.8	0.6	2.2	2.0	1.9	2.2	2.2	2.2
Inflation (quarterly % change, a.r.)													
Consumer Price Index				3.1	2.5	3.6	6.1	1.5	2.4	2.4	1.5	1.7	2.5
CPI ex Food & Energy				3.2	2.0	2.8	2.9	2.0	2.5	3.1	1.9	2.2	2.8
PCE Price Index				2.8	2.9	4.6	5.3	2.2	2.1	2.6	1.7	1.8	2.3
PCE ex Food & Energy				2.9	2.7	4.4	3.6	2.6	2.1	3.0	1.9	2.1	2.4
Inflation (4-quarter % change)													
Consumer Price Index	2.7	3.4	2.1	2.9	2.7	2.7	3.8	3.4	3.4	3.1	2.0	2.0	2.1
CPI ex Food & Energy	2.7	2.5	2.5	3.1	2.7	2.5	2.7	2.4	2.5	2.6	2.3	2.4	2.5
PCE Price Index	2.8	3.5	2.1	2.7	2.8	3.1	3.9	3.7	3.5	3.0	2.2	2.1	2.1
PCE ex Food & Energy	2.9	3.2	2.4	2.9	2.9	3.1	3.4	3.3	3.2	2.8	2.4	2.3	2.4
Monetary Policy													
Fed Funds Target (% midpoint of target range)	3.625	3.625	3.125	4.125	3.625	3.625	3.625	3.625	3.625	3.375	3.125	3.125	3.125
Fiscal Policy													
Federal Budget balance (% of GDP)	-5.4	-6.1	-5.7										

1) GDP less contribution from inventory investment

2) GDP less contributions from inventory investment and trade

3) GDP less contributions from inventory investment, trade, and the government sector. (Private final consumption plus investment.)

Source: Bureau of Economic Analysis, Bureau of Labor Statistics, Federal Reserve, Census Bureau, Treasury Dep't, Morgan Stanley Research forecasts

Exhibit 27: Monthly inflation forecasts

	% Change - Year-over-Year				% Change - Month-over-Month				Headline CPI NSA Index
	Headline PCE	Core PCE	Headline CPI	Core CPI	Headline PCE	Core PCE	Headline CPI	Core CPI	
Jan-25	2.6	2.7	3.0	3.3	Jan-25	0.32	0.28	0.43	317.671
Feb-25	2.6	2.9	2.8	3.1	Feb-25	0.38	0.42	0.23	319.082
Mar-25	2.3	2.6	2.4	2.8	Mar-25	0.00	0.08	0.03	319.799
Apr-25	2.2	2.5	2.3	2.8	Apr-25	0.19	0.22	0.16	320.795
May-25	2.4	2.7	2.4	2.8	May-25	0.22	0.27	0.10	321.465
Jun-25	2.6	2.9	2.7	2.9	Jun-25	0.32	0.31	0.25	322.561
Jul-25	2.7	2.9	2.7	3.1	Jul-25	0.16	0.23	0.23	323.048
Aug-25	2.7	2.9	2.9	3.1	Aug-25	0.22	0.18	0.35	323.976
Sep-25	2.8	2.8	3.0	3.0	Sep-25	0.27	0.20	0.30	324.800
Oct-25	2.7	2.8	2.9	2.8	Oct-25	0.21	0.24	0.13	#N/A
Nov-25	2.8	2.8	2.7	2.6	Nov-25	0.18	0.14	0.13	324.122
Dec-25	2.9	2.9	2.7	2.6	Dec-25	0.33	0.33	0.30	324.054
Jan-26	2.8	3.1	2.4	2.5	Jan-26	0.32	0.41	0.17	325.252
Feb-26	2.8	3.0	2.4	2.5	Feb-26	0.33	0.32	0.27	326.785
Mar-26	3.5	3.2	3.3	2.6	Mar-26	0.66	0.29	0.87	330.213
Apr-26	3.6	3.2	3.8	2.7	Apr-26	0.36	0.19	0.64	333.020
May-26	4.0	3.3	4.2	2.8	May-26	0.54	0.42	0.47	335.123
Jun-26	3.6	3.2	3.5	2.6	Jun-26	-0.07	0.17	-0.42	333.952
Jul-26	3.6	3.2	3.3	2.5	Jul-26	0.16	0.25	0.07	333.918
Aug-26	3.7	3.2	3.4	2.4	Aug-26	0.34	0.24	0.41	335.077
Sep-26	3.8	3.2	3.7	2.4	Sep-26	0.38	0.17	0.58	336.869
Oct-26	3.7	3.1	3.5	2.5	Oct-26	0.08	0.18	0.02	336.382
Nov-26	3.6	3.2	3.5	2.6	Nov-26	0.13	0.17	0.13	335.798
Dec-26	3.5	3.0	3.5	2.6	Dec-26	0.19	0.17	0.23	335.512
Jan-27	3.4	3.0	3.5	2.6	Jan-27	0.30	0.37	0.23	336.952
Feb-27	3.3	2.9	3.4	2.6	Feb-27	0.16	0.21	0.14	338.104
Mar-27	2.8	2.7	2.8	2.6	Mar-27	0.20	0.16	0.23	339.514
Apr-27	2.5	2.7	2.2	2.4	Apr-27	0.11	0.16	0.07	340.474
May-27	2.1	2.4	1.8	2.3	May-27	0.12	0.15	0.08	341.294
Jun-27	2.3	2.4	2.3	2.4	Jun-27	0.12	0.15	0.10	341.869
Jul-27	2.3	2.3	2.3	2.4	Jul-27	0.12	0.18	0.08	341.853
Aug-27	2.1	2.3	2.1	2.4	Aug-27	0.19	0.20	0.21	342.378
Sep-27	2.0	2.3	1.8	2.4	Sep-27	0.22	0.20	0.27	343.172
Oct-27	2.0	2.3	1.9	2.5	Oct-27	0.13	0.20	0.10	342.974
Nov-27	2.1	2.4	2.0	2.5	Nov-27	0.19	0.20	0.22	342.695
Dec-27	2.1	2.4	2.1	2.5	Dec-27	0.25	0.20	0.32	342.697

Source: BLS, BEA, Morgan Stanley Research forecasts. PCE numbers include the impact of PCE historical revisions.

US Outlook Scenarios

Exhibit 28: Morgan Stanley US Economics baseline and alternative outlooks for the US Economy: 2026, and 2027

	Aggregate Demand Shock: Animal Spirits (20%)	AI Productivity Boost with Labor Displacement (10%)	De-escalation: Mild Energy Headwind (45%)	Permanent Oil Premium Regime (15%)	Global Oil-Led Recession (10%)
Key Forecasts					
GDP growth	The oil shock abates. Elevated wealth supports consumption and "animal spirits" keep business spending elevated. GDP grows 2.9% in 2026 and 3.1% in 2027, 0.5-1.0pp faster than the baseline.	GDP growth is steady, rising 2.2% 4Q/4Q in 2026 and 2.1% in 2027. Strong business spending is offset somewhat by weaker labor markets and low consumer confidence. Despite softer inflation in 2027, consumer spending stays sluggish as labor displacement mounts.	High oil prices act as a tax on consumption, neutralizing stimulus from higher tax refunds in the OBBA. AI capex spending remains robust. Real growth held back to 2.3% (4Q/4Q) in 2026 before rebounding to 2.6% in 2027.	Oil prices average \$120/bbl in 2026 and \$100/bbl in 2027. Strategic stockpiling occurs amid a repricing of geopolitical risk in oil logistics. Mild demand destruction occurs. Real GDP grows in 3Q and 4Q 2026. Real GDP rises 2.0% in 2026 (4Q/4Q) before a 2.4% rebound in 2027.	Oil prices average \$140-160/bbl through 3Q and end the year at \$100/bbl. Supply shortages ensue, and demand destruction occurs. The result: mild recession with two quarters of negative GDP growth in 3Q and 4Q 2026. Real GDP rises only 0.1% in 2026 (4Q/4Q) before a 2.5% rebound in 2027.
Labor markets	Stronger demand results in tighter labor markets. The unemployment rate falls to 3.9% in 2026 and 3.5% in 2027.	The economy experiences a prolonged period of net zero job creation. The separation rate edges higher, but widespread layoffs are avoided. The unemployment rate creeps higher, rising to 4.8% in 2027, 0.8pp above our baseline outlook.	The "curious balance" continues. Average monthly payroll gains slow to 43k this year and firm modestly to 60k next year. Labor demand remains soft on geopolitical uncertainty. Less immigration means a lower breakeven rate of hiring. The unemployment rate finishes 2026 at 4.3% and 2027 at 4.1%.	The "curious balance" continues. Average monthly payroll gains slow to 30k this year and firm modestly to 50k next year. Labor demand remains soft on geopolitical uncertainty. Less immigration means a lower breakeven rate of hiring. The unemployment rate finishes 2026 at 4.3% and 2027 at 4.2%.	Payrolls fall an average of 30k per month in 2026 with declines near 200k in Q4. In 2027, payrolls rebound, but bring the unemployment rate down slowly. The unemployment rate rises to 5.5% this year and 5.3% next year.
Inflation	Even with the oil shock dissipating, inflation pressures step-up. Headline inflation is 3.7% in 2026 and 2.7% in 2027. Core PCE inflation firms through Q1 '27, prompting Fed hikes, which subsequently slow inflation.	Inflation rises on account of higher energy prices, but weak labor markets and slow wage growth provide offset. Headline inflation peaks at 3.8% y/y in 2Q 2026 – similar to the baseline – and finishes the year at 3.3%. Inflation falls back to the 2.0% target in 2027.	High inflation in 2026 dissipates in 2027. Energy prices push headline PCE inflation to 3.2% 4Q/4Q in 2026. Headline and core PCE decelerate to 2.0% and 2.3% 4Q/4Q in 2027 as the oil shock fades. Energy is a headline story with no second-round effects on core.	Two more years of above-target inflation. Energy prices push headline PCE inflation to 3.8% 4Q/4Q in 2026, while core moves sideways at 3.1% on tariff push and second round effects from the oil shock. Headline and core PCE decelerate to 2.2% and 2.8% 4Q/4Q in 2027 as the oil shock fades.	Headline inflation surges to 5.1% y/y in 3Q 2026 and 4.5% at year-end on elevated oil prices. On weaker demand, inflation decelerates to 1.1% in 2027.
Federal reserve policy	Hikes to counter inflation. The Fed raises rates by 100bp beginning in 4Q 2026 to 4.50-4.75%, more than reversing the 75bp in pre-emptive cuts to support the labor market in 2025.	Labor displacement leads to cuts. Above-target inflation keeps the Fed on the sidelines in 2026, but a rising unemployment rate leads the Fed to cut by 75bp in 2027 to a target range of 2.75-3.0%.	Normalization continues in 2027. Above-target inflation keeps the Fed on the sidelines in 2026, but falling y/y rates of inflation toward 2.0% leads the Fed to cut by 50bp in 2027 to a target range of 3.0-3.25%.	A prolonged pause. Inflation persistence means a high bar for cuts. Tariff effects must prove transitory, and energy-driven inflation must show a clear reversal before cuts are considered. The Fed maintains the target range of 3.50-3.75% through the end of 2027.	The Fed shifts to cuts as downside risks to growth mount. 200bp in 2026 and a terminal of 1.50-1.75%. 100bp of hikes in 2027 for a target range of 2.50-2.75% in 4Q 2027. No QE.
Key Forecasts					
Consumer spending	Better than expected equity market returns and a rebound in labor markets boost consumption to around 3.0% in 2026 and back to near 2.0% in 2027. Upper-income consumers drive much of the spending, but nominal wage growth supports low- and middle-income spending.	Consumption rises 1.6% in 2026 and 1.5% in 2026. High gas prices weaken goods consumption in 2026. Soft labor market conditions and fears of AI displacement make consumers turn cautious in 2027 despite improving real spending power from decelerating inflation.	Consumption slows to 1.8% in 2026 as high gasoline prices weigh on goods spending. A resilient economy supports wealth effects from upper income consumers. In 2027, consumption rebounds to 2.1% 4Q/4Q as inflation decelerates.	Consumption slows to 1.4% in 2026 as high gasoline prices weigh on goods spending. In 2027, consumption rebounds to 2.2% 4Q/4Q as inflation decelerates and a recession is avoided.	Consumption falls from 3Q 2026 to 4Q 2026 before a gradual reacceleration. Households cut back on durable goods spending. Negative wealth effects lead upper-income consumers to pull back, with spillover into broader spending.
Nonresidential fixed investment	Business spending broadens out beyond AI, due to animal spirits and fiscal support.	Similar to the baseline. AI-related spending remains robust, leading nonresidential fixed investment to rise 7.0% in 2026 and 8.0% in 2027. AI adoption rates rise and firms realize productivity gains	AI-related spending remains robust, leading nonresidential fixed investment to rise 7.0% in 2026 and 8.0% in 2027. Sentiment around AI capex is not dented by geopolitical concerns.	AI-related spending remains strong, leading nonresidential fixed investment to rise 6.0% in 2026 and 7.0% in 2027. Sentiment is somewhat dented by geopolitical concerns and elevated energy costs.	A retrenching business sector cuts employment and capex. Investment declines throughout 2H and into Q1 2027, followed by recovery.
Residential investment	Stronger growth is balanced by higher rates in 2026, resulting in still limited housing activity over the forecast horizon.	Housing activity remains sluggish in both 2026 and 2027, despite lower interest rates. Improved affordability is offset by softening labor market conditions.	Housing remains "stuck" due to low affordability and limited inventory. Residential investment falls 1.3% in 2026 and rises 1.5% in 2027.	Housing remains "stuck" due to low affordability and limited inventory. High inflation and no Fed cuts mean higher mortgage rates. Residential investment falls 1.8% in 2026 and is flat in 2027.	Housing activity is weak in 2026 on the back of slower growth and weak labor markets. Activity picks up in 2027 as growth recovers and low rates spur demand.
Net trade	Imports are strong with strong demand, resulting in a larger drag on GDP and a wider trade deficit.	Relative to the baseline, imports are softer. AI-related imports are robust, but imports of consumer goods are more muted on the back of softer household spending. Net trade subtracts 0.2pp from growth in 2026 and 2027.	Imports are strong with strong demand, thus a larger drag on GDP. Imports rise 6.3% in 2026 and 5.0% in 2027. Exports rise 4.1% and 1.8%. After adding 0.4pp to growth in 2025, net trade subtracts 0.4pp from growth in 2026 and 2027.	Imports rise 5.3% in 2026 and 4.0% in 2027. Exports rise 3.0% and 1.0%. Net trade subtracts 0.4pp from growth in 2026 and 2027.	Imports and exports both fall over the next few quarters as demand falls and slow global growth limits export demand. Net trade contributes positively to growth in 2026.
Fiscal policy	Fiscal multipliers are higher than in the baseline, resulting in higher contribution to growth. With stronger growth, fiscal deficits decline. The deficit as a percent of GDP is 5.7% in 2026 and 5.6% in 2027.	Similar to the baseline. A softer labor market means slightly less revenue growth and more automatic stabilizers. The deficit is 6.3% of GDP in 2026 and 2027.	The fiscal impulse in the OBBA is neutralized by higher gasoline prices. The midterm elections bring divided government, leading to budget gridlock and large deficits at about 6.0% of GDP	Higher gas prices more than offset the growth effects of the OBBA. The midterm elections bring divided government, leading to budget gridlock and large deficits at about 6.3% of GDP	Deficits rise further as tax receipts decelerate and automatic stabilizers boost transfer payments. Defense spending picks up. The deficit as a percent of GDP is 6.9% in 2026 and 6.7% in 2027
Credit conditions	Credit conditions loosen somewhat on the heels of stronger growth and animal spirits. Bank lending accelerates.	Credit conditions bifurcate, with willingness to lend to consumers worsening amid solid growth in C&I lending.	The pace of tightening in lending standards has slowed. Geopolitical uncertainty keeps conditions tighter over the course of the year, though we see modest easing over the forecast horizon.	Geopolitical uncertainty keeps conditions modestly tighter over the forecast horizon.	Credit conditions tighten further as the economy contracts. By late this year, the Fed has eased policy enough to encourage some new credit supply but demand remains weak.
Productivity growth	Similar productivity growth as the baseline, even with stronger employment growth.	Similar productivity growth as the baseline, only with greater labor displacement.	Productivity rises 2.1% in 2027. AI capex keeps output per hour elevated in high-tech industries. Productivity gains are from higher output, not less employment.	Productivity rises 1.7% in 2027. Less robust AI-related spending limits productivity growth in the short-run.	With rapid declines in output, productivity falls over the next few quarters. Productivity then begins to recover in 2027.
Consumer and business confidence	Confidence rebounds across the course of the year, and further in 2027.	Labor displacement pushes consumer confidence down further, testing new lows. Business optimism remains elevated.	Confidence of households and business remains near 2022-2023 lows on account of low labor demand, high inflation, and elevated uncertainty.	Confidence of households and business remains near 2022-2023 lows on account of low labor demand, high inflation, and elevated uncertainty.	Recession pushes confidence down further, testing new lows.

Source: Morgan Stanley Research forecasts

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